

JATINJAYASIMHA VN

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SUMMARY

Cybersecurity M.Tech student with hands-on application security and AppSec automation experience, including production AI-assisted vulnerability detection and a self-built DevSecOps CI/CD security pipeline on Azure. Strong fundamentals in OWASP Top 10, cloud security (AWS/Azure IAM, misconfigurations), and security tool integration. Seeking an AppSec engineering role to apply offensive/defensive testing skills and contribute to secure SDLC practices.

EDUCATION

Master of Technology in Cyber Security

2024 – Present

BMS Institute of Technology and Management • Bengaluru, Karnataka • CGPA 9.14

Bachelor of Technology in Electronics and Communication Engineering

2018 – 2022

Reva University • Bengaluru, Karnataka • CGPA 8.83

SKILLS

Programming: Java, Python, C | **Networking & Security:** TCP/IP, DNS, HTTP/S, Snort, Suricata, OSSEC, SIEM, IDS/IPS, Firewalls, ELK Stack

Security Tools: Kali Linux, Burp Suite, Nmap, Wireshark, Metasploit, Splunk, Nessus, OSINT, VMware

Cloud & DevSecOps: AWS (IAM, S3, security groups), Azure, Terraform, Ansible, GitLab CI/CD, Docker, Trivy, Semgrep, Gitleaks, OWASP Dependency Check

Auth & Identity: OAuth 2.0, Firebase Auth, RBAC | **Tools:** Git, GitHub, MySQL

Core Concepts: Application/Network/Cloud Security, MITRE ATT&CK, OWASP Top 10, CVSS

Compliance: NIST, PCI-DSS, ISO/IEC 27001, GDPR, HIPAA

EXPERIENCE

Software Engineer Trainee – Application Security

Aug 2022 – Aug 2023

Cognizant Technology Solutions, Bengaluru, Karnataka

- Developed an enterprise security remediation platform integrating Azure OpenAI for intelligent code analysis, reducing manual review time by ~40% and accelerating secure development workflows.
- Mapped detected vulnerabilities to 200+ CWE classifications and automated OWASP-aligned remediation outputs, increasing consistency and accuracy of vulnerability reporting by ~30%; outputs directly supported audit evidence pipelines.
- Designed prompt workflows and built REST APIs for AI-assisted vulnerability detection and secure code generation, enabling automated fixes for ~60–70% of common security findings reducing mean time to remediate (MTTR).

Application Development and Management Intern

Mar 2022 – Jul 2022

Cognizant Technology Solutions, Bengaluru, Karnataka

- Built a React.js/Express.js/Node.js/MongoDB UI wrapper for Jenkins with APIs to trigger jobs, sync GitHub data, and run test suites with real-time interaction.

PROJECTS

Automated Secure Software Delivery Pipeline using DevSecOps Toolchain on Azure

- Provisioned a complete Azure environment (resource group, VNet, subnet, network security group, public IP, Linux VM) using Terraform IaC, debugging real free-tier capacity and SKU restrictions to keep infrastructure reproducible and destroyable on demand.
- Automated VM configuration with Ansible (Docker installation, service setup, user permissions), resolving an Ansible–Python/OS compatibility conflict that required rebuilding on a newer Ubuntu image — real-world infra debugging beyond tutorial scope.
- Built a 4-stage GitLab CI/CD pipeline (Build → Test → Security → Deploy) triggered on every push, integrating Trivy (container/filesystem scanning), Semgrep (SAST), Gitleaks (secret detection), and OWASP Dependency Check (SCA) as parallel security gates.
- Automated zero-touch deployment over SSH (container teardown, code sync, rebuild, restart) with no manual intervention; resolved SSH key format (RSA vs ED25519) and CI/CD variable-scoping issues end-to-end.

Cloud-Native Application Suite — Codec Technologies Internship

- Built and deployed six live cloud apps across a 3-month program — real-time systems (Firebase chat, live resource-monitoring dashboard), cloud storage (AWS S3 + MongoDB Atlas), and a Cloud Log Monitoring/Alert mini-SIEM (Firestore ingestion, error-spike alerts ≥3/60s, Chart.js analytics) comparable to AWS CloudWatch/ELK Stack.

- Implemented cloud-native security and access control: Firebase Auth (Google OAuth 2.0), role-based Firestore rules, admin-only panels; practiced IAM policy design, cost monitoring, multi-platform deployment (Vercel, Render, GitHub Pages).

Optimized Real-Time VAPT Framework for Zero-Day Exploit Detection — PG Major

Oct 2025 – Present

- Set up a two-device Kali Linux environment (attacker and detection module) over a local network to simulate attack scenarios including port scanning, brute-force attempts, and denial-of-service traffic; captured synchronized network and system telemetry to build a labeled dataset.
- Applied Isolation Forest for unsupervised anomaly detection, optimized using Genetic Algorithms for feature selection and Particle Swarm Optimization for hyperparameter tuning; integrated a Fuzzy Inference System (FIS) for human-interpretable risk classification of detected anomalies.
- Validated detection models against CIC-IDS and UNSW-NB15 benchmark datasets; produced evaluation reports with accuracy metrics, comparison plots, and CSV exports to document detection performance.

API Security Assessment using ML and Simulated VAPT Techniques

Jun 2025 – Aug 2025

- Created an intentionally vulnerable REST API (Flask + SQLite) to simulate real-world security flaws for penetration testing and dataset generation.
- Built automated request logging, feature extraction, and an ML detection pipeline (Isolation Forest, optional Random Forest), with end-to-end CSV/PNG reporting to demonstrate detection results.

CERTIFICATIONS

- **Ethical Hacker** — Cisco Networking Academy, 2025
- **Digital Forensics Masterclass (Complete Computer Forensics)** — EC-Council, 2025
- **Java Programming** — Frontlines Media
- **Python Programming** — Python Bootcamp, Udemy